cs-techrep
Technical Reports in Computer Science and Software Engineering

Christoph P. Neumann <cyberpetaneuron@gmail.com>

Version 0.9.1

Abstract

The `cs-techrep` class provides a convenient environment for writing technical reports in computer science and software engineering. The style is adopted from IEEE. The cs-techrep class requires the latex/pdflatex in combination with biblatex using a biber backend.

1 Installation

The `zip` or `tar.gz` file comes with a `cs-techrep.ins` and a `cs-techrep.dtx` file included which contains the LaTeX stuff.

To extract the class files call:

```
$ latex cs-techrep.ins
```

This call will extract all LaTeX specific files to the current directory. You can either use the files for a single cv project or you can integrate the files into your TeX installation.

If you just want to use cs-techrep for a single curriculum vitae project, the simplest way is just to copy the generated files to the folder of the project.

If you want to integrate cs-techrep into your TeX installation, create a directory `tex/latex/cs-techrep` beneath your TeX installation (e.g. beneath `/usr/share/texmf`) and copy all files from the current directory there. Now call:

```
$ mktexlsr
```

to update the file–cache of LaTeX.

Hint: The cs-techrep distribution contains a sample docstrip configuration in `docstrip.cfg` via which files can be distributed automatically to their correct positions inside a LaTeX installation. Feel free to adapt this file to your environment and afterwards call `latex cs-techrep.ins` to install the package to its right place.

2 Templates

For a quick start the iaria-lite distribution contains document templates. The templates can be found in the `cs-techrep-example-ctan.zip` file.

3 Documentclass

`documentclass cs-techrep` This package provides the documentclass `cs-techrep`. The documentclass supports the following options:

- `conference` Passed to IEEEtran
- `a4paper` Passed to IEEEtran
- `subfig` Loads subfig package with custom style settings
- `subcaption` Loads subcaption package with custom style settings
- `flushend` Activate flushend package (compatible with arXiv build process)
- `pbalance` Activate pbalance package (incompatible with arXiv build process)

A Remark about two-column document balancing on the last page: The flushend package is recommended, because it works within the arXiv automated build process. However, flushend has a major incampatibility with package lineno, which is, e. g., transitively loaded by package mindflow. Thus, in case when flushend does not have any effect, check whether one of your packages loads lineno. Usually you are stuck with your packages and, thus, will instead be forced to switch from flushend to pbalance. Please be aware that pbalance works great and has high compatibility, but unfortunately it will not have any effect within the arXiv automated build process. In case of both, a somehow needed lineno package and an intended arXiv upload, I recommend to do without two-column balancing on the last page and just to stay away from both flushend and pbalance, in order to ensure that your paper is layouted identically in any initial publication and a later arXiv upload. I hope this remark proves helpful, it took me some nerve to find out.

There is also another important difference between flushend and pbalance: the handling of footnotes on the last page. Flushend provides a decent result, but the result of pbalance is incomprehensible. I strongly recommend avoiding footnotes on the last page. If you absolutely need footnotes on the last page, consider staying away from two-column balancing on the last page.

About subfigures: Both well-known packages subfig and subcaption can be used. However, they are not compatible with each other and we can either load one or the other. The document class provides options to load them for you with some customized style settings.

4 Requirements

We instrument several other LaTeX packages for different purposes, which must be available under your installation.

- IEEEtran
- acronym
- afterpage
- babel
- biblatex
- biblatex-ieee
- booktabs
- ccicons
- cleveref
- csquotes
- diagbox
- doclicense
- etoolbox
- extdash
- flushend
- fontenc
- graphicx
- hypcap
- hyperref
- hyperref
- inputenc
- lipsum
- listings
- lmodern
- lscape
- microtype
- newtxtt

- orcidlink
- paralist
- pbalance
- pdflscape
- siunitx
- stfloats
- subcaption
- subfig
- tcolorbox
- times
- todonotes
- upquote
- url
- xcolor
- xpatch
- xspace

5 The Code

```
@echo off

rem ******************************************************************************
rem * author: Christoph P. Neumann
rem ******************************************************************************
cd /d "%~dp0"

echo == ..\examples\* ==
for /D %%D in (..\examples\*) do (
 echo == %%D ==
 xcopy /Y ".\*.cls" "%%~D"
)

pause
```

```
%# Copyright (c) 2023-2026 Christoph P. Neumann,
%# Copyright (c) 2018-2021 Oliver Kopp
%# Reference: https://latextemplates.github.io/

% !TeX spellcheck = en-US
% LTeX: language=en-US
% !TeX encoding = utf8
```

```
% !TeX program = pdflatex
% !BIB program = biblatex / biber

\NeedsTeXFormat{LaTeX2e}

\ProvidesClass{cs-techrep}[2026/02/24 TechReps in CS and SWE v0.9.1]

% Define marker counter for options (counter are global in latex):

\newcounter{cstechrep@subfigtracker}
\newcounter{cstechrep@subcaptiontracker}
\newcounter{cstechrep@pbalancetracker}
\newcounter{cstechrep@flushendtracker}

% Declare options:

\DeclareOption{subfig}{
  \setcounter{cstechrep@subfigtracker}{1}
  \typeout{cs-techrep class: subfig option is set (use subfloat command).}
}
\DeclareOption{subcaption}{
  \setcounter{cstechrep@subcaptiontracker}{1}
  \typeout{cs-techrep class: subcaption option is set (use subfigure command).}
}

% IEEE option onecolumn will be ignored
\DeclareOption{onecolumn}{\OptionNotUsed}

\DeclareOption{pbalance}{
  \setcounter{cstechrep@pbalancetracker}{1}
  \typeout{cs-techrep class: pbalance option is set.}
}
\DeclareOption{flushend}{
  \setcounter{cstechrep@flushendtracker}{1}
  \typeout{cs-techrep class: flushend option is set.}
}

% We'll pass any document class options along to the underlying class
% (recommended IEEE options: conference,a4paper)
\DeclareOption*{%
  \PassOptionsToClass{\CurrentOption}{IEEEtran}
}

% Now we'll execute any options passed in
\ProcessOptions\relax

% Instead of defining each and every little detail required to create a new
% document class, you can base your class on an existing document class.
\LoadClass{IEEEtran}

% This template has been tested with IEEEtran of 2015.

% ========================
% USER PROVIDED VARIABLES
```

```
76
77 \RequirePackage{etoolbox}
78
79 \makeatletter
80 \newcommand*\DefcstechrepVar[1]{%
81  \@namedef{cstechrep#1}{}% initialize
82  \@namedef{cstechrepSet#1}##1{%
83   \expandafter\gdef\csname cstechrep#1\endcsname{##1}}%
84 }
85 \makeatother
86
87 \DefcstechrepVar{AuthorListTex} % Use a comma ',' as author separator!
88 \DefcstechrepVar{AuthorListBib} % Use the word "and" as author separator!
89 \DefcstechrepVar{TitleTex}
90 \DefcstechrepVar{TitleBib}
91 \DefcstechrepVar{Department}  % e.g., Department of Computer Science
92 \DefcstechrepVar{Institution} % e.g., OTH Amberg-Weiden
93 \DefcstechrepVar{Address}     % e.g., Amberg, Germany
94 \DefcstechrepVar{Type}        % e.g., Technical Report
95 \DefcstechrepVar{Year}        % e.g., 2025
96 \DefcstechrepVar{Month}       % e.g., 10 (for October)
97 \DefcstechrepVar{Number}      % e.g., CL-TR-2024-42
98 \DefcstechrepVar{Lang}        % e.g., english = en-US (or: ngerman = DE)
99
100 %%% Default values (can be overriden):
101 \cstechrepSetAuthorListTex{Vorname1 Nachname1, Vorname2 Nachname2, Vorname3 Nachname3, Vornam
102 \cstechrepSetAuthorListBib{Vorname1 Nachname1 and Vorname2 Nachname2 and Vorname3 Nachname3 a
103 % Capitalization: https://capitalizemytitle.com/style/Chicago/
104 \cstechrepSetTitleTex{FancyName: The Cloud-Native Foo for Bar}
105 \cstechrepSetTitleBib{{\cstechrepTitleTex}}
106 \cstechrepSetDepartment{CyberLytics\-/Lab at the Department of Electrical Engineering, Media,
107 \cstechrepSetInstitution{Ostbayerische Technische Hochschule Amberg\-/Weiden}
108 \cstechrepSetAddress{Amberg, Germany}
109 \cstechrepSetType{Technical Report}
110 \cstechrepSetNumber{TR-\cstechrepYear-99}
111 \cstechrepSetLang{english}
112
113 % ======================
114
115 % backticks (`) are rendered as such in verbatim environments.
116 % See following links for details:
117 %   - https://tex.stackexchange.com/a/341057/9075
118 %   - https://tex.stackexchange.com/a/47451/9075
119 %   - https://tex.stackexchange.com/a/166791/9075
120 \RequirePackage{upquote}
121
122 % Set English as language and allow to write hyphenated"=words
123 %
124 % Even though `american`, `english` and `usenglish` are synonyms
125 % for babel package (according to
126 % https://tex.stackexchange.com/questions/12775/babel-english-american-usenglish),
127 % the llncs document class is prepared to avoid the overriding of certain
128 % names (such as "Abstract." -> "Abstract" or "Fig." -> "Figure")
129 % when using `english`, but not when using the other 2.
```

```
% english has to go last to set it as default language
\RequirePackage[ngerman,main=english]{babel}
%
% Hint by http://tex.stackexchange.com/a/321066/9075 -> enable "= as dashes
\addto\extrasenglish{\languageshorthands{ngerman}\useshorthands{"}}

% CM variant with scalable font
% (to prevent "pdfTeX error (font expansion):
% auto expansion is only possible with scalable")
\RequirePackage{lmodern}
% Times New Roman, because IEEE is based on Times (and not CM/lmodern),
% however, package times is not a scalable font, thus, lmodern is required
% additionally and must be loaded before times.
\RequirePackage{times}

% Links behave as they should. Enables "\url{...}" for URL typesettings.
% Allow URL breaks also at a hyphen, even though it might be confusing:
% Is the "-" part of the address or just a hyphen?
% See https://tex.stackexchange.com/a/3034/9075.
\RequirePackage[hyphens]{url}

% When activated, use text font as url font, not the monospaced one.
% For all options see https://tex.stackexchange.com/a/261435/9075.
% \urlstyle{same}

% Improve wrapping of URLs - hint by http://tex.stackexchange.com/a/10419/9075
\makeatletter
\g@addto@macro{\UrlBreaks}{\UrlOrds}
\makeatother

% nicer // - solution by http://tex.stackexchange.com/a/98470/9075
% DO NOT ACTIVATE -> prevents line breaks
%\makeatletter
%\def\Url@twoslashes{\mathchar`\/\@ifnextchar/{\kern-.2em}{}}
%\g@addto@macro\UrlSpecials{\do\/{\Url@twoslashes}}
%\makeatother

% use nicer font for code
\RequirePackage[zerostyle=b,scaled=.75]{newtxtt}

% inputenc must be loaded before biblatex
\RequirePackage[utf8]{inputenc}

% Has to be loaded AFTER any font packages.
% See https://tex.stackexchange.com/a/2869/9075.
\RequirePackage[T1]{fontenc}


% Character protrusion and font expansion.
% See http://www.ctan.org/tex-archive/macros/latex/contrib/microtype/
\RequirePackage[
  babel=true, % Enable language-specific kerning
  expansion=alltext,
  protrusion=alltext-nott, % Ensure no changes at the edge of the listing.
```

```
  nopatch=eqnum, % fix unable to apply patch eqnum
  final % Always enable microtype, even if in draft mode.
        % This helps finding bad boxes quickly. In the standard
        % configuration, this template is always in the final mode, so
        % this option only makes a difference if "pros" use the draft mode.
]{microtype}

% \texttt{test -- test} keeps the "--" as "--"
% (and does not convert it to an en dash)
\DisableLigatures{encoding = T1, family = tt* }

\RequirePackage{graphicx}

% Creative Commons
\RequirePackage{ccicons}

% Diagonal lines in a table
% http://tex.stackexchange.com/questions/17745/diagonal-lines-in-table-cell
% Slashbox is not available in texlive (due to licensing)
% and also gives bad results. Thus, we use diagbox
\RequirePackage{diagbox}

\RequirePackage[cmyk, table]{xcolor}

\RequirePackage{relsize} % allows for \textsmaller{..}

\RequirePackage[nolist,smaller]{acronym}

% Code Listings
\RequirePackage{listings}

\definecolor{eclipseStrings}{RGB}{42,0.0,255}
\definecolor{eclipseKeywords}{RGB}{127,0,85}
\colorlet{numb}{magenta!60!black}

% JSON definition
% Source: https://tex.stackexchange.com/a/433961/9075

\lstdefinelanguage{json}{
  basicstyle=\normalfont\ttfamily,
  commentstyle=\color{eclipseStrings}, % style of comment
  stringstyle=\color{eclipseKeywords}, % style of strings
  numbers=left,
  numberstyle=\scriptsize,
  stepnumber=1,
  numbersep=8pt,
  showstringspaces=false,
  breaklines=true,
  frame=lines,
  % backgroundcolor=\color{gray}, %only if you like
  string=[s]{"}{"},
  comment=[l]{:\ "},
  morecomment=[l]{:"},
  literate=
```

```
      *{0}{{{\color{numb}0}}}{1}
       {1}{{{\color{numb}1}}}{1}
       {2}{{{\color{numb}2}}}{1}
       {3}{{{\color{numb}3}}}{1}
       {4}{{{\color{numb}4}}}{1}
       {5}{{{\color{numb}5}}}{1}
       {6}{{{\color{numb}6}}}{1}
       {7}{{{\color{numb}7}}}{1}
       {8}{{{\color{numb}8}}}{1}
       {9}{{{\color{numb}9}}}{1}
}

\lstset{
  % everything between (* *) is a latex command
  escapeinside={(*}{*)},
  %
  language=json,
  %
  showstringspaces=false,
  %
  extendedchars=true,
  %
  basicstyle=\footnotesize\ttfamily,
  %
  commentstyle=\slshape,
  %
  % default: \rmfamily
  stringstyle=\ttfamily,
  %
  breaklines=true,
  %
  breakatwhitespace=true,
  %
  % alternative: fixed
  columns=flexible,
  %
  numbers=left,
  %
  numberstyle=\tiny,
  %
  basewidth=.5em,
  %
  xleftmargin=.5cm,
  %
  % aboveskip=0mm,
  %
  % belowskip=0mm,
  %
  captionpos=b
}

% Enable Umlauts when using \lstinputputlisting.
% See https://stackoverflow.com/a/29260603/873282 für details.
% listingsutf8 did not work in June 2020.
```

```
\lstset{literate=
  {á}{{\'a}}1 {é}{{\'e}}1 {í}{{\'i}}1 {ó}{{\'o}}1 {ú}{{\'u}}1
  {Á}{{\'A}}1 {É}{{\'E}}1 {Í}{{\'I}}1 {Ó}{{\'O}}1 {Ú}{{\'U}}1
  {à}{{\`a}}1 {è}{{\`e}}1 {ì}{{\`i}}1 {ò}{{\`o}}1 {ù}{{\`u}}1
  {À}{{\`A}}1 {È}{{\'E}}1 {Ì}{{\`I}}1 {Ò}{{\`O}}1 {Ù}{{\`U}}1
  {ä}{{\"a}}1 {ë}{{\"e}}1 {ï}{{\"i}}1 {ö}{{\"o}}1 {ü}{{\"u}}1
  {Ä}{{\"a}}1 {Ë}{{\"E}}1 {Ï}{{\"I}}1 {Ö}{{\"o}}1 {Ü}{{\"U}}1
  {â}{{\^a}}1 {ê}{{\^e}}1 {î}{{\^i}}1 {ô}{{\^o}}1 {û}{{\^u}}1
  {Â}{{\^A}}1 {Ê}{{\^E}}1 {Î}{{\^I}}1 {Ô}{{\^O}}1 {Û}{{\^U}}1
  {Ã}{{\~A}}1 {ã}{{\~a}}1 {Õ}{{\~O}}1 {õ}{{\~o}}1
  {œ}{{\oe}}1 {Œ}{{\OE}}1 {æ}{{\ae}}1 {Æ}{{\AE}}1 {ß}{{\ss}}1
  {ů}{{\H{u}}}1 {Ů}{{\H{U}}}1 {ő}{{\H{o}}}1 {Ő}{{\H{O}}}1
  {ç}{{\c c}}1 {Ç}{{\c C}}1 {ø}{{\o}}1 {å}{{\r a}}1 {Å}{{\r A}}1
}

% Allows for column type X
% In tabularx environment, e.g., \begin{tabularx}{\textwidth}{lX}
\RequirePackage{tabularx}
% Definition of booktabs commands (\toprule, \midrule, \bottomrule)
% without booktabs package:
% (However, the smallskip still breaks vertical lines!)
\newcommand{\toprule}{\hline\noalign{\smallskip}}
\newcommand{\midrule}{\noalign{\smallskip}\hline\noalign{\smallskip}}
\newcommand{\bottomrule}{\noalign{\smallskip}\hline}

% Extended enumerate, such as \begin{compactenum}
\RequirePackage[alwaysadjust]{paralist}

% Read: Is there a disadvantage to using natbib=true with biblatex?
% https://tex.stackexchange.com/questions/149313
% /is-there-a-disadvantage-to-using-natbib-true-with-biblatex
% Substitutions:
% Natbib \textcite = biblatex \textcite
% Natbib \citep = biblatex \parencite

\RequirePackage[
    defernumbers=true,  % required due to self-cite
    %style=ieee,        % english only
    sorting=none,       % nty, nyt, nyvt, anyt, anyvt, ynt, ydnt, none, debug
    sortcites=false,
    citestyle=numeric-comp, % BEWARE: ieee-comp would improve IEEE-style-compatibility, but i
    bibstyle=numeric,
    backend=biber,
    bibencoding=utf8,   % = auto, utf8, ascii
    hyperref=true,      % activate hyperref support
    backref=false,      % activate backrefs
    isbn=true,
    url=true,
    doi=true,
    urldate=iso,        % = iso, iso8601, year, short, long, terse (terse is like iso/iso8601
    seconds=true,       % iso requires also seconds=true
    maxnames=5,         % max number of names in text ??
    minnames=1,         % min number of names in text ??
    maxbibnames=99,     % max number of names in bibliography
```

```
    minbibnames=1,      % number of names in bibliography (bevor et al)
    maxcitenames=99,    % max number of names in text ??
    mincitenames=1,     % max number of names in text ??
    giveninits=false,   % abbreviate first names
    uniquename=init,    % otherwise option conflict with firstinits=true
    autolang=other
]{biblatex}

% Due to applied option defernumbers=true:
% Prevent citations in the references are being numbered as '0' in ArXiv
% https://tex.stackexchange.com/questions/463556
% /all-the-citations-in-the-references-are-being-numbered-as-0-in-arxiv-while-upl
\makeatletter
\let\blx@rerun@biber\relax
\makeatother

% selfref.bib needs to be generated, containing a bibentry with key selfref
\addbibresource{selfref.bib}
\DeclareBibliographyCategory{selfref}
\addtocategory{selfref}{selfref}
% How to use: \printbibliography[notcategory=selfref]

% Same fontsize as without natbib
\renewcommand{\bibfont}{\normalfont\small}

% Zotero generates wrong language fields with wrong "en"
% instead of correct "english" option, thus, despite
% default biblatex clearlang, the "en" will incorrectly
% be printed out, even in english papers.
% Make "en" equivalent to "english" and/or its derivatives:
\DeclareRedundantLanguages{en}{english,american,british,
  canadian,australian,newzealand,USenglish,UKenglish}

% Enable hyperlinked author names in the case of \textcite
% Source: https://tex.stackexchange.com/a/76075/9075
\RequirePackage{etoolbox}
\makeatletter
\patchcmd{\NAT@test}{\else \NAT@nm}{\else \NAT@hyper@{\NAT@nm}}{}{}
\makeatother

\makeatletter
\DeclareCiteCommand{\citemonth}
  {\usebibmacro{prenote}}
  {%
    \let\oldabx@bibmonth\abx@bibmonth%
    \setkeys{blx@opt@pre}{dateabbrev=false}% nowork
    \mkbibmonth{\thefield{month}}%
    \let\abx@bibmonth\oldabx@bibmonth% restore
  }
  {\multicitedelim}
  {\usebibmacro{postnote}}
\makeatother

%% Enable nice TODO comments
```

```
\RequirePackage[colorinlistoftodos,prependcaption,textsize=tiny]{todonotes}
\newcommand{\TODO}[1]{\todo[%
linecolor=red,backgroundcolor=red!25,bordercolor=red]{#1}}

% bring the table caption in the same line as label "Table I" (with etoolbox)
\makeatletter
\patchcmd{\@makecaption}
  {\\}
  {.\ }
  {}
  {}
\makeatother

% For controlling the placement of the captions
% (table: top | figure: bottom), the package caption
% recommends to use package floatrow:
% (Without foatrom, the positioning is determined by
%  the textual (!) position of command \label in relation to
%  includegraphics/begin-tabular; a well-known latex aspect.)
\RequirePackage{floatrow}
\floatsetup[table]{capposition=top,captionskip=5pt}
\floatsetup[figure]{capposition=bottom,captionskip=8pt}

% Put footnotes below floats
% Source: https://tex.stackexchange.com/a/32993/9075
% and allow floats in a twocolumn document with [t] or [b]
\RequirePackage{stfloats}
\fnbelowfloat

\RequirePackage[group-minimum-digits=4,per-mode=fraction]{siunitx}
\addto\extrasgerman{\sisetup{locale = DE}}

% Enable that parameters of \cref{}, \ref{}, \cite{}, ...
% are linked so that a reader can click on the number
% and jump to the target in the document
\RequirePackage{hyperref}

% Enable hyperref without colors and without bookmarks
\hypersetup{
  hidelinks,
  colorlinks=true,
  allcolors=black,
  pdfstartview=Fit,
  breaklinks=true
}

% Enable correct jumping to figures when referencing
\RequirePackage[all]{hypcap}

\RequirePackage{orcidlink}

% For easy quotations: \enquote{text}
% This package is very smart when nesting is applied,
% otherwise textcmds (see below) provides a shorter command
```

```
\RequirePackage[autostyle=true]{csquotes}

% Enable using "‘quote"’
% See: https://tex.stackexchange.com/a/150954/9075
\defineshorthand{"‘}{\openautoquote}
\defineshorthand{"’}{\closeautoquote}

% amsmath must be loaded before cleveref
\usepackage{amsmath,amssymb,amsfonts}

% Extensions for references inside the document (\cref{fig:sample}, ...)
% Enable usage \cref{...} and \Cref{...} instead of \ref:
% Type of reference included in the link.
% That means, "Figure 5" is a full link instead of just "5".
% !! Underlying rules for "capitalize" and "noabbrev": APA style
% http://www.sagepub.com/upm-data/27160_APA_Style.pdf
% Use a capital letter "S" when saying "In Section 2, we..."
% Don’t use capital letters when saying "There are four sections, ..."
% Capitalize when figure is followed by a number, e.g., "In Figure 3, we...."
% Don’t use when no number follows, e.g., "The two figures ..."
% (Please notice: CMOS would prefer lower letters, in contrast to APA.)
\RequirePackage[capitalise,nameinlink,noabbrev]{cleveref}

\crefname{listing}{Listing}{Listings}
\Crefname{listing}{Listing}{Listings}
\crefname{lstlisting}{Listing}{Listings}
\Crefname{lstlisting}{Listing}{Listings}

\RequirePackage{lipsum}

\RequirePackage{tcolorbox}
\tcbuselibrary{listings}

%introduce \powerset
% See: http://matheplanet.com/matheplanet/nuke/
% html/viewtopic.php?topic=136492&post_id=997377
\DeclareFontFamily{U}{MnSymbolC}{}
\DeclareSymbolFont{MnSyC}{U}{MnSymbolC}{m}{n}
\DeclareFontShape{U}{MnSymbolC}{m}{n}{
  <-6>    MnSymbolC5
  <6-7>   MnSymbolC6
  <7-8>   MnSymbolC7
  <8-9>   MnSymbolC8
  <9-10>  MnSymbolC9
  <10-12> MnSymbolC10
  <12->   MnSymbolC12%
}{}
\DeclareMathSymbol{\powerset}{\mathord}{MnSyC}{180}

\RequirePackage{xspace}
%\newcommand{\eg}{e.\,g.,\xspace}
%\newcommand{\ie}{i.\,e.,\xspace}
\newcommand{\eg}{e.\,g.,\ }
\newcommand{\ie}{i.\,e.,\ }
```

```
508
509 % Enable hyphenation at other places as the dash.
510 % Example: application\hydash specific
511 \makeatletter
512 \newcommand{\hydash}{\penalty\@M-\hskip\z@skip}
513 % Definition of "=
514 % See: http://mirror.ctan.org/macros/latex
515 % /contrib/babel-contrib/german/ngermanb.dtx
516 \makeatother
517
518 % Enable copy and paste of text from the PDF
519 % Only required for pdflatex. It "just works" in the case of lualatex.
520 \input glyphtounicode
521 \pdfgentounicode=1
522
523 \RequirePackage{lscape}
524 \RequirePackage{pdflscape}
525 \RequirePackage{afterpage}
526
527 % requires, e.g., \PassOptionsToPackage{
528 % type=CC,modifier=by,version=4.0}{doclicense} before \documentclass
529 \RequirePackage[hyperxmp=false,lang=English]{doclicense}
530
531 % Improve wrapping of URLs
532 % See: http://tex.stackexchange.com/a/10419/9075
533 \makeatletter
534 \def\ps@IEEEtitlepagestyle{
535   \def\@oddfoot{\mycopyrightnotice}
536   \def\@evenfoot{}
537 }
538 \def\mycopyrightnotice{
539   {\footnotesize
540     \begin{minipage}{0.8\textwidth}
541   \centering
542     % License based on doclicense package:
543     \href{\doclicenseURL}{\doclicenseIcon{}
544     This work is licensed under a \doclicenseLongNameRef{}
545     (\doclicenseNameRef{}) license.}
546     % Citation advice:
547     \\Please cite as: \fullcite{selfref}.
548   \end{minipage}
549   }
550 }
551 \makeatother
552
553 \makeatletter
554 % allows for forcing command expansion inside a filecontents* environment
555 % (cf. https://tex.stackexchange.com/questions/534035
556 % /macros-commands-inside-a-filecontents-environment-does-not-expand)
557 \def\filecontentsForceExpansion#1#2#3{
558   \global\let\ltxspecials\dospecials
559   \gdef\dospecials{\ltxspecials
560     \catcode`#1=0
561     \catcode`#2=1
```

```
    \catcode`#3=2
    \global\let\dospecials\ltxspecials
  }
}
\makeatother

\title{\cstechrepTitleTex}

\author{
\IEEEauthorblockN{\cstechrepAuthorListTex}
\IEEEauthorblockA{%
\cstechrepDepartment\\
\cstechrepInstitution\\
\cstechrepAddress
}}

\IEEEspecialpapernotice{\cstechrepType: \cstechrepNumber,
\citemonth{selfref} \cstechrepYear}

% subfigure captions with parentheses and footnotesize:
\ifnum\value{cstechrep@subfigtracker}=1
  \typeout{cs-techrep class: load subfig package (use subfloat command).}
  \RequirePackage[caption=false,font=footnotesize]{subfig}
\fi
\ifnum\value{cstechrep@subcaptiontracker}=1
  \typeout{cs-techrep class: load subcaption package (use subfigure command).}
  \RequirePackage{subcaption}
  \captionsetup[subfigure]{labelformat=parens,font=footnotesize}
\fi

% Balance/level columns at the last page / in bibliography
% DISCLAIMER: No solution is a 100%/perfect one!
% Mainstream option: pbalance (see https://ctan.org/pkg/pbalance)
% BUT: incompatible with arxiv automated process (!)
\ifnum\value{cstechrep@pbalancetracker}=1
  \typeout{cs-techrep class: load pbalance package.}
  \RequirePackage{pbalance}
\fi
% Flushend => works with arxiv, but major incampatibility with lineno
% (and, thus, also with mindflow et al.)
\ifnum\value{cstechrep@flushendtracker}=1
  \typeout{cs-techrep class: load flushend package.}
  \RequirePackage{flushend}
\fi

% ========================================================
% Packages that need to be loaded last:

% Allows to hyphenate a word that contains a dash:
% https://stackoverflow.com/questions/2193307
% /how-do-i-get-latex-to-hyphenate-a-word-that-contains-a-dash
% Use \-/ for a breakable dash that does not prevent the remainer
% of the word to be hyphenated.
\RequirePackage[shortcuts]{extdash}
```

```
% ========================================================

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
% this file. LaTeX will ignore anything after this line.
\endinput
```